

GUANGHUI (DANIEL) LIU, PhD

COMPUTATIONAL BIOLOGY & BIOINFORMATICS SCIENTIST | BIOMEDICAL AI | REPRODUCIBLE RESEARCH SYSTEMS

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PROFESSIONAL PROFILE

Computational scientist with 10+ years of experience turning high-dimensional biomedical data into **validated models, biological candidates, reproducible computational workflows, and publication-ready evidence**. I combine omics analysis, statistical and machine learning methods, software architecture, and translational interpretation. My distinguishing strength is building the research system around the analysis - not just running isolated scripts - so data states, parameters, models, evidence, and downstream conclusions remain traceable, testable, and reusable.

SELECTED ACHIEVEMENTS

- Built a full-stack reproducible omics research system spanning data/metadata QC, feature discovery, model training, interpretation, external validation, evidence synthesis, and report/manuscript-oriented output; designed for both cloud and local execution.
- Designed modular algorithm operations with standardized adapters and versioned execution, enabling methods to be added, configured, tested, enabled/disabled, and replaced without breaking downstream workflows.
- Implemented leakage-resistant model development with fold-local feature selection, repeated cross-validation, untouched holdout evaluation, model/result fingerprints, dependency invalidation, and persistent analysis provenance.
- Created a model registry that preserves every successfully trained algorithm as a versioned candidate or winner artifact with selected genes, feature importance, CV metrics, run identity, and reproducibility metadata - making model comparison auditable rather than winner-only.
- Developed translational IBD models using ~18,989-gene transcriptomic data plus clinical covariates; improved treatment-response prediction AUC from 0.71 to 0.82 while producing interpretable gene/risk summaries for biomarker follow-up.
- Led breast-cancer multi-omics survival research integrating gene expression, DNA methylation, clinical covariates, and time-to-event outcomes; corresponding author of a Bioinformatics publication.

SIGNATURE TECHNICAL ACHIEVEMENT - REPRODUCIBLE OMICS RESEARCH PLATFORM

Designed and built an end-to-end research platform that converts fragmented analysis scripts into a controlled, auditable data-to-evidence workflow. **The system demonstrates the computational infrastructure I can bring to research teams: reproducibility, extensibility, validation discipline, and fast translation from analysis to scientific output.**

- Persistent analysis sessions retain cutoffs, feature counts, CV folds, random seeds, selected genes, trained-model artifacts, metrics, plots, interpretation tables, and upstream/downstream relationships across iterative analyses.
- Algorithm registry/adaptor architecture separates method-specific execution from standardized result contracts; current workflows support feature ranking/selection and predictive modeling methods including mRMR, ReliefF, LASSO/Elastic Net, Random Forest, and XGBoost, with a framework designed for continued expansion.
- External-validation workflows connect trained signatures/models to independent cohorts through metadata harmonization, reproducibility checks, ROC/PR-AUC/C-index/calibration, SHAP/feature contribution, pathway/enrichment context, and candidate-level evidence summaries.
- Research-delivery layer converts analysis artifacts into consistent figures, tables, methods/results summaries, reproducibility records, reports, and manuscript-oriented outputs, reducing the gap between exploratory computation and PI/reviewer-ready evidence.
- Local-first execution mode can keep raw biomedical data and intermediate/model artifacts on the researcher workstation while retaining the same structured workflow and analysis provenance used in cloud execution.

TRANSLATIONAL RESEARCH HIGHLIGHTS

Inflammatory Bowel Disease - Transcriptomics, Prediction & Gene-Signature Discovery

- Integrated high-dimensional transcriptomics with PUCAI/Mayo scores, histology, demographics, and treatment-response outcomes; benchmarked LSTM, Random Forest, XGBoost, LASSO, and StepAIC approaches and developed AUC-driven reinforcement-learning feature selection.

Breast Cancer - Multi-Omics Survival & Biomarker Modeling

- Integrated mRNA expression, DNA methylation, clinical covariates, and survival outcomes; used lmQCM-derived eigengenes and adaptive multi-task learning combining Cox and ordinal objectives for patient stratification and biomarker-oriented interpretation.

PROFESSIONAL EXPERIENCE

Project Scientist / Postdoctoral Scientist | Cedars-Sinai Medical Center

Los Angeles, CA

2023-2026

- Led translational ML and biomarker-discovery analyses using high-dimensional transcriptomic and clinical data; owned the workflow from preprocessing and feature ranking through cross-validation, interpretation, visualization, and scientific communication.
- Developed reproducible Python/R research workflows and mentored junior researchers in statistical modeling, computational organization, troubleshooting, and presentation of results to biomedical collaborators.

Research Associate, Biomedical Informatics | Intelligent Bio Systems Laboratory, SUNY Oswego

Oswego, NY

2019-2022

- Led funded computational precision-oncology research integrating molecular and clinical data for breast-cancer survival prediction, multi-omics biomarker analysis, and clinically interpretable risk stratification.
- Developed adaptive multi-task survival models and reusable Python/R workflows; translated methodological work into peer-reviewed publications and reproducible research outputs.

Assistant Professor, Information Engineering | Nanjing University of Finance and Economics

Nanjing, China

2010-2018

- Conducted and supervised machine-learning research in bioinformatics and biomedical imaging, including protein-protein interaction-site prediction, protein subcellular localization, feature engineering, imbalanced learning, and ensemble modeling.
- Taught and mentored students in machine learning, data mining, programming, quantitative research, and scientific communication.

CORE TECHNICAL CAPABILITIES

Omics / Bioinformatics: RNA-seq and gene-expression analysis, DNA methylation, multi-omics integration, survival data, biomarker/signature discovery, pathway/enrichment analysis, public genomic repositories, metadata harmonization; working familiarity with Seurat and single-cell workflow concepts.

Statistical / ML Methods: mRMR, ReliefF, LASSO/Elastic Net, Random Forest, XGBoost, LSTM, autoencoders, Cox models, reinforcement-learning feature selection, repeated cross-validation, ROC/PR-AUC, C-index, calibration, SHAP and feature-importance analysis.

Programming / Scientific Computing: Python, R, Bash, Linux/Unix, C/C++, Java, Perl, SQL; pandas, NumPy, scikit-learn, PyTorch, TensorFlow/Keras, Bioconductor, DESeq2, edgeR, clusterProfiler, ImQCM, Cytoscape, Jupyter.

Research Software / Pipelines: Git, Docker, conda, Snakemake, Nextflow, MLflow, HPC-style execution, FastAPI, React, SQLite/MySQL; modular algorithm registries/adapters, standardized data contracts, parameter/provenance persistence, model artifact management, runtime/smoke testing, dependency invalidation, automated reporting.

Research Delivery: Study design support, reproducibility planning, publication-quality figures/tables, technical documentation, methods/results drafting, investigator-facing interpretation, mentoring, and translation of computational findings into testable biological hypotheses.

SELECTED PUBLICATIONS

- Liu G. An Explainable Deep Network Framework with Case-based Reasoning Strategies for Survival Analysis in Cancer. Scientific Reports. 2026; accepted, in publication process.
- Bichindaritz I, Liu G (corresponding), Bartlett C. Integrative survival analysis of breast cancer with gene expression and DNA methylation data. Bioinformatics. 2021;37(17):2601-2608.
- Liu G-H, Zhang B-W, Qian G, et al. Bioimage-based prediction of protein subcellular location in human tissue with ensemble features and deep networks. IEEE/ACM Transactions on Computational Biology and Bioinformatics. 2020;17(6):1966-1980.
- Liu G-H, Shen H-B, Yu D-J. Prediction of protein-protein interaction sites with machine-learning-based data-cleaning and post-filtering procedures. Journal of Membrane Biology. 2016;249(1-2):141-153.
- Bichindaritz I, Liu G (corresponding). Adaptive multi-omics survival analysis in cancer. KES-InMed. 2022.

EDUCATION

PhD: Control Engineering & Artificial Intelligence - Nanjing University of Science & Technology, China (2011-2017)

MS: Computer Control Engineering - Jiangnan University, China (2007-2010)

BS: Electrical Automation - Jiangnan University, China (2002-2006)

RESEARCH LEADERSHIP, FUNDING & SERVICE

Funding: SUNY Performance Fund Grant (EIPF #172), Main Researcher; National Natural Science Foundation of China (No. 71701089), Main Researcher; Natural Science Foundation of Jiangsu Province (08KJD520016), Key Researcher.

Scientific Service: Associate Editor, PLOS ONE; reviewer for Bioinformatics, Bioinformatics Advances, IEEE/ACM Transactions on Computational Biology and Bioinformatics, and related computational/biomedical venues.